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| <p style="text-align: center;"><b>OPERATING SYSTEMS [3 0 0 3]</b></p> <p style="text-align: center;"><b>SEMESTER – V</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |                   |                       |
| <b>Subject Code</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CSF 3125  | <b>IA Marks</b>   | <b>50</b>             |
| <b>Number of Lecture Hours/Week</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>03</b> | <b>Exam Marks</b> | <b>50</b>             |
| <b>Total Number of Lecture Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>36</b> | <b>Exam Hours</b> | <b>03</b>             |
| <b>CREDITS – 03</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |           |                   |                       |
| <p><b>Course objectives:</b> This course will enable students to</p> <ul style="list-style-type: none"> <li>• Understand the fundamental concepts of the basic privileges, structure and process scheduling of the operating system.</li> <li>• Comprehend the concepts of deadlock, memory management, virtual memory and disk structure concepts.</li> <li>• Know the concepts of the file system and protection of operating system.</li> </ul>                                                                                                                                                                           |           |                   |                       |
| <b>Module -1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |                   | <b>Teaching Hours</b> |
| <p><b>INTRODUCTION:</b></p> <p>What Operating Systems Do, Operating System Structure, Operating System Operations, Process Management, Memory Management, Storage Management.</p> <p><b>OPERATING SYSTEM STRUCTURE:</b></p> <p>Operating System Services, User and Operating System Interface, System Calls, Types of System Calls, System Programs, Operating System Structure, Virtual Machines, System Boot.</p> <p><b>PROCESS CONCEPT:</b></p> <p>Overview, Process Scheduling, Operations on Processes, Interprocess Communication.</p> <p><b>THREADS:</b></p> <p>Overview, Multithreaded Models, Thread Libraries.</p> |           |                   | <b>9 Hours</b>        |

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| <p><b>Text Book 1:</b> Chapter: 1: 1.1,1.4-1.8, Chapter: 2: 2.1– 2.5, 2.6,2.7.1-2.7.3, 2.10, Chapter: 3: 3.1-3.4, Chapter: 4: 4.2, 4.3</p>                                                                                                                                                                                                                                                                                                                                                                                 |                        |
| <p><b>Module -2</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                        |
| <p><b>PROCESS SCHEDULING:</b></p> <p>Basic Concepts, Scheduling Criteria, Scheduling Algorithms, Thread Scheduling</p> <p><b>SYNCHRONIZATION:</b></p> <p>Background, The Critical Section Problem, Peterson’s Solution, Synchronization Hardware, Mutex Locks, Semaphores.</p> <p><b>DEADLOCKS:</b></p> <p>System Model, Deadlock, Characterization, Methods for handling deadlocks, Deadlock prevention, Deadlock Avoidance.</p> <p><b>Text Book 1:</b> Chapter 5: 5.1-5.4, Chapter: 6: 6.1-6.6, Chapter: 7: 7.1 -7.5</p> | <p><b>11 Hours</b></p> |
| <p><b>Module – 3</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                        |
| <p><b>MEMORY MANAGEMENT STRATEGIES:</b></p> <p>Logical Versus Physical Address Space, Swapping, Contiguous Memory Allocation, Segmentation, Paging, Structure of the Page Table.</p> <p><b>VIRTUAL-MEMORY MANAGEMENT:</b></p> <p>Background, Demand Paging, Copy-On-Write, Page Replacement, Allocation of Frames, Thrashing.</p> <p><b>MASS-STORAGE STRUCTURE:</b></p> <p>Disk Structure, Disk Scheduling, Swap-Space Management</p>                                                                                      | <p><b>10 Hours</b></p> |

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| <b>Text Book 1:</b> Chapter 8: 8.1.3,8.2-8.6, Chapter 9: 9.1-9.6, Chapter 12: 12.2 12.4,12.6                                                                                                                                                                                                                                                                                                                                                                                                        |                |
| <b>Module - 4</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |
| <b>FILE-SYSTEM:</b><br><br>File Concept, Access Methods, Directory and Disk Structure, File System Mounting<br><br><b>SYSTEM PROTECTION:</b><br>Goals of Protection, Principles of Protection, Domain of Protection, Access Matrix Implementation of Access Matrix.<br><br><b>Text Book 1:</b> Chapter 10: 10.1- 10.4, Chapter 14 :14.1- 14.5                                                                                                                                                       | <b>6 Hours</b> |
| <b>Course outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                |
| After studying this course, students will be able to: <ol style="list-style-type: none"> <li>1. Summarize the operating system functions, services and structure.</li> <li>2. Compare the various process scheduling and disk scheduling algorithms</li> <li>3. Apply the various methods for synchronization and deadlock prevention</li> <li>4. Analyse the various memory management techniques</li> <li>5. Explain the concepts of file system and protection</li> </ol>                        |                |
| <b>Text Books:</b> <ol style="list-style-type: none"> <li>1. A. Silberschatz, P. B. Galvin and G. Gagne, <i>Operating System Concepts</i>, (9e), Wiley and Sons (Asia) Pte Ltd, 2016.</li> </ol>                                                                                                                                                                                                                                                                                                    |                |
| <b>Reference Books:</b> <ol style="list-style-type: none"> <li>1. D.M. Dhamdhere, <i>Operating Systems A Concept based Approach</i>,(2e), Tata McGraw-Hill 2006</li> <li>2. Deitel &amp; Deitel , <i>Operating systems</i>, (3e), Pearson Education, India. 2018.</li> <li>3. <u>Neil Matthew</u>, <u>Richard Stones</u>, <i>Beginning Linux Programing</i>(4e), Wiley, 2007</li> <li>4. Andrew S Tanenbaum, Herbert Bos, <i>Modern Operating systems</i>. Pearson Education India, 2018</li> </ol> |                |